E commerce store dataset

Project Objective

To clean, analyze, and visualize an E-commerce sales dataset using **Excel formulas, Pivot Tables, and charts** to derive meaningful business insights.

Dataset Description

Dataset: E-commerce Sales
Common Columns:

- Order Date
- Customer Name
- Customer_ID
- Sales_ID
- Brand
- Stock
- Sub Category
- Category
- Product Name
- Quantity
- Unit_Price
- Discount
- Total_Sales
- Region
- Payment Mode

STEP 1: Data Import & Organization

Step 1.1: Import Data

- Open Excel → **Data → Get Data → From Workbook**
- Load data into a new sheet named **Raw Data**

1.Data cleaning process (with excel formulas)

- Remove Duplicates
- Select entire table
- Data->Remove Duplicate

2. Handle Missing values:

Missing Unit_Price

Formula =IF(G2="",AVERAGE($G$2:$G$2001),G2)

Missing Category

Formula =IF(C2="","Unknown",C2)

3. Standardize Date Format:

Formula =Text(B2,"dd-mm-yyyy")

4.Create Total Sales Column

Create a new column in excel (Total_Sales)

Formula =E2*F2

Total_Sales
218.04
313.41
3656.2
921.51
2235.9
441.03
1570.62
1369.56
448.25
1746.34
278.63
1147.68
2578.32
633.8
1613.88
871.34
903.96
1005.38

Step2 Data Transformation – combine tables using vlookup function

Fetch Customer Name:

Formula=VLOOKUP(C2,CUSTOMER!A:B,2,FALSE)

Fetch City:

Formula=VLOOKUP(C2,CUSTOMER!A:E,5,FALSE)

Fetch Product Name

Formula=VLOOKUP(D2,Product_Dim!A:C,2,FALSE)

Fetch Category

Formula=VLOOKUP(D2,Product_Dim!A:C,3,FALSE)

Fetch Region

Formula==VLOOKUP(E2,Store_Dim!A:E,3,FALSE)

Step3. Data Analysis for toolpak:

Total_Sales
$218.04
$313.41
$3,656.20
$921.51
$2,235.90
$441.03
$1,570.62
$1,369.56
$448.25
$1,746.34
$278.63
$1,147.68
$2,578.32
$633.80
$1,613.88
$871.34
$903.96
$1,005.38
$1,118.46

Tookpak	
Mean	$1,276.21
Standard Error	$21.25
Median	$1,008.27
Mode	$1,541.16
Standard Deviation	$950.03
Sample Variance	$902,562.51
Kurtosis	-$0.13
Skewness	$0.82
Range	$3,971.66
Minimum	$21.14
Maximum	$3,992.80
Sum	$2,551,141.87
Count	$1,999.00

Step4. Pivot table Analysis:

Pivot table 1: Category-wise sales

- **Rows->Category**
- **Values->Sum of sales**

Total_Sale	Month	Year	Category	Region
$218.04	July	2024	Sports	North
$313.41	Decembe	2023	Electronic	North
$3,656.20	June	2024	Beauty	West
$921.51	Novembe	2023	Electronic	West
$2,235.90	January	2024	Beauty	North
$441.03	July	2024	Electronic	South
$1,570.62	July	2025	Home	South
$1,369.56	Novembe	2023	Home	West
$448.25	May	2025	Home	North
$1,746.34	July	2025	Home	East
$278.63	February	2024	Sports	South
$1,147.68	May	2025	Electronic	North
$2,578.32	August	2024	Sports	South
$633.80	August	2025	Electronic	West
$1,613.88	June	2025	Sports	North
$871.34	Decembe	2024	Clothing	South
$903.96	Septembe	2025	Sports	East
$1,005.38	Decembe	2024	Home	South
$1,118.46	August	2024	Electronic	South

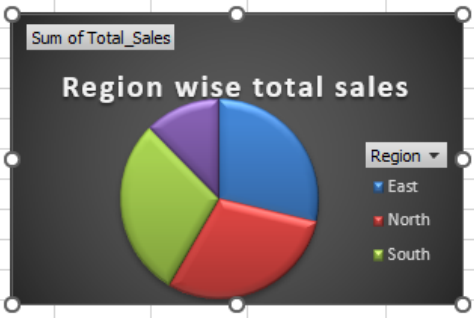
Category	Sum of Total_Sales
Beauty	$381,570.69
Clothing	$441,510.13
Electronics	$593,630.43
Home	$499,136.10
Sports	$635,512.56
Grand Total	**$2,551,359.91**

Pivot table 2: Region-wise sales

- **Rows->Region**
- **Values->Sum of Total_Sales**

Total_Sale	Month	Year	Category	Region
$218.04	July	2024	Sports	North
$313.41	December	2023	Electronic	North
$3,656.20	June	2024	Beauty	West
$921.51	Novembe	2023	Electronic	West
$2,235.90	January	2024	Beauty	North
$441.03	July	2024	Electronic	South
$1,570.62	July	2025	Home	South
$1,369.56	Novembe	2023	Home	West
$448.25	May	2025	Home	North
$1,746.34	July	2025	Home	East
$278.63	February	2024	Sports	South
$1,147.68	May	2025	Electronic	North
$2,578.32	August	2024	Sports	South
$633.80	August	2025	Electronic	West
$1,613.88	June	2025	Sports	North
$871.34	December	2024	Clothing	South
$903.96	Septembe	2025	Sports	East
$1,005.38	December	2024	Home	South
$1,118.46	August	2024	Electronic	South

Region	Sum of Total_Sales
East	$732,884.19
North	$757,560.42
South	$744,313.97
West	$316,601.33
Grand Total	**$2,551,359.91**



Pivot table 3: Monthly Sales

- **Rows->Order_Date(Group->Month)**
- **Values->Average of Total_Sales**

Order_Date	Month	Total_Sales
07-07-2024	July	218.04
25-12-2023	December	313.41
03-06-2024	June	3656.2
29-11-2023	November	921.51
06-01-2024	January	2235.9
26-07-2024	July	441.03
12-07-2025	July	1570.62
18-11-2023	November	1369.56
04-05-2025	May	448.25
31-07-2025	July	1746.34
13-02-2024	February	278.63
24-05-2025	May	1147.68
24-08-2024	August	2578.32
06-08-2025	August	633.8
21-06-2025	June	1613.88
10-12-2024	December	871.34
13-09-2025	September	903.96
11-12-2024	December	1005.38
13-08-2024	August	1118.46

Month	Average of Total_Sales
2023	**$1,215.3**
Qtr4	**$1,215.3**
Oct	$1,162.8
Nov	$1,139.6
Dec	$1,320.6
2024	**$1,285.7**
2025	**$1,279.6**
Grand Total	**$1,275.7**

Pivot table 3: Monthly Sales

- **Rows->Order_Date(Group->Month)**
- **Values->Sum of Total_Sales**

Order_Date	Month	Total_Sales
07-07-2024	July	218.04
25-12-2023	December	313.41
03-06-2024	June	3656.2
29-11-2023	November	921.51
06-01-2024	January	2235.9
26-07-2024	July	441.03
12-07-2025	July	1570.62
18-11-2023	November	1369.56
04-05-2025	May	448.25
31-07-2025	July	1746.34
13-02-2024	February	278.63
24-05-2025	May	1147.68
24-08-2024	August	2578.32
06-08-2025	August	633.8
21-06-2025	June	1613.88
10-12-2024	December	871.34
13-09-2025	September	903.96
11-12-2024	December	1005.38
13-08-2024	August	1118.46

Month	Sum of Total_Sales
2023	$263,725.3
Qtr4	$263,725.3
Oct	$52,327.0
Nov	$99,143.4
Dec	$112,254.9
2024	$1,284,445.2
2025	$1,003,189.5
Grand Total	$2,551,359.9

Step5.Data visualization (Charts)

1.Column Chart- Sales by Category

- Source : Pivot Table
- Chart Type: Clustered Column

2. Pie Chart – Region-wise Sales

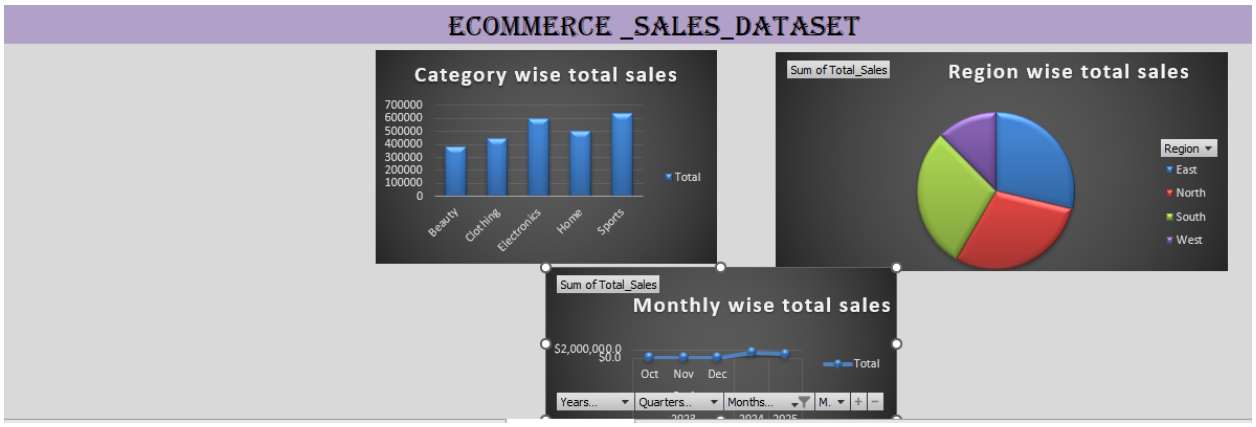
- Shows % contribution by region

3.Line Chart – Monthly Sales Trend

- Helps identify peak sales months

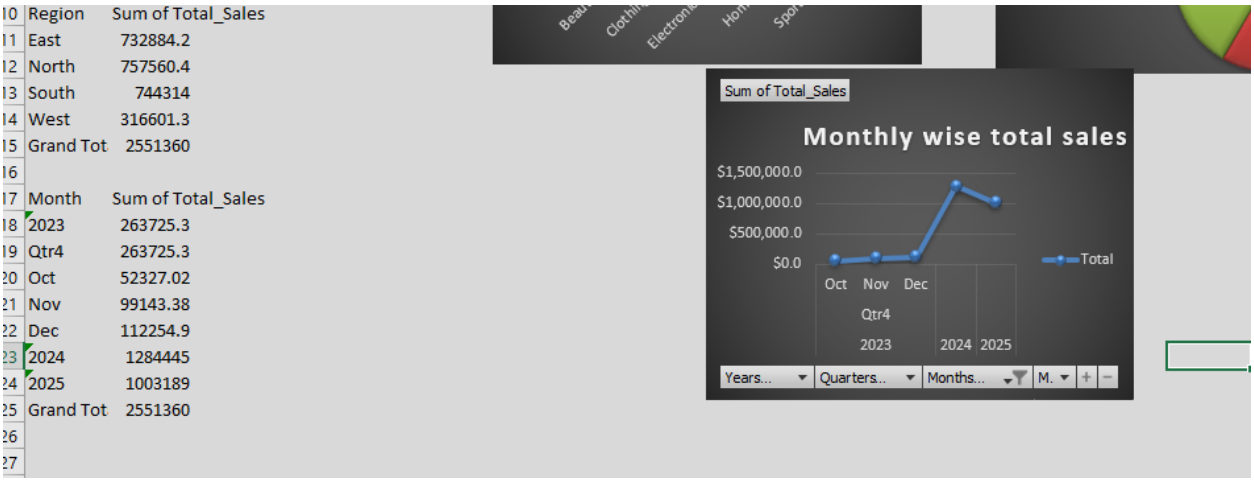
Step6.Dashboard Components:

Category, region monthly wise Total_Sales



Category	Sum of Total_Sales
Beauty	381570.7
Clothing	441510.1
Electronic	593630.4
Home	499136.1
Sports	635512.6
Grand Tot	2551360

Region	Sum of Total_Sales
East	732884.2
North	757560.4
South	744314
West	316601.3
Grand Tot	2551360

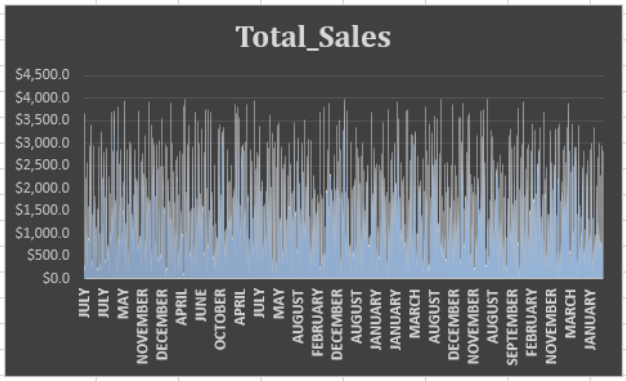


Region	Sum of Total_Sales
East	732884.2
North	757560.4
South	744314
West	316601.3
Grand Tot	2551360

Month	Sum of Total_Sales
2023	263725.3
Qtr4	263725.3
Oct	52327.02
Nov	99143.38
Dec	112254.9
2024	1284445
2025	1003189
Grand Tot	2551360

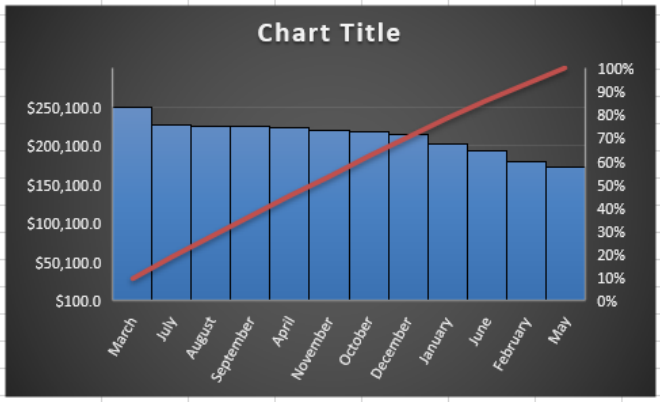
Step7.Forecast of trend in sales

Month	Total_Sales
July	$218.0
Decembe	$313.4
June	$3,656.2
Novembe	$921.5
January	$2,235.9
July	$441.0
July	$1,570.6
Novembe	$1,369.6
May	$448.3
July	$1,746.3
February	$278.6
May	$1,147.7
August	$2,578.3
August	$633.8
June	$1,613.9
Decembe	$871.3
Septembe	$904.0
Decembe	$1,005.4
August	$1,118.5



- Sales peak during festive months for trendline

Month	Total_Sales
July	$218.0
Decembe	$313.4
June	$3,656.2
Novembe	$921.5
January	$2,235.9
July	$441.0
July	$1,570.6
Novembe	$1,369.6
May	$448.3
July	$1,746.3
February	$278.6
May	$1,147.7
August	$2,578.3
August	$633.8
June	$1,613.9
Decembe	$871.3
Septembe	$904.0
Decembe	$1,005.4
August	$1,118.5



1. Descriptive Insight (What happened?)

It explains past data.

Shows summary like total, count, average.

Example:

Total sales were ₹5,00,000 and 1,200 orders were made.

2. Diagnostic Insight (Why did it happen?)

It finds the reason behind the result.

Uses comparison and analysis.

Example:

Sales decreased because there were fewer discounts.

3. Predictive Insight (What will happen?)

It predicts the future using past data.

Based on trends and patterns.

Example:

Sales may increase during festival season.

4. Prescriptive Insight (What should we do?)

It gives suggestions or actions.

Helps improve results.

Example:

Offer discounts to increase sales.

Key Business Insights

- Electronics category generates the highest revenue
- South region contributes maximum sales
- Sales peak during festive months
- Few product contribute to majority of revenue

Conclusion:

This excel capstone project demonstrates end-to-end data analytics including:

- Data cleaning
- Table combination
- Excel formulas

- Pivot analysis
- Business visualization

It reflects real-world e-commerce analysis and decision making skills.